

Build a Multi-Agent System Prototype

Scenario

You're the technical lead for a startup developing an intelligent research assistant that helps academic researchers discover, analyze, and synthesize information from multiple sources. The system needs to handle complex research queries that require different types of expertise—literature search, data analysis, citation verification, and synthesis.

Your Challenge

Build a working multi-agent system prototype that demonstrates effective task decomposition, inter-agent communication, and controlled autonomy.

Your Task

Create a functional multi-agent system prototype using one of the frameworks discussed in the course (CrewAI, AutoGen, or LangGraph). Your system should demonstrate the key principles of multi-agent architecture, communication, and governance.

Requirements

1. System Architecture

- Implement at least 3 specialized agents with distinct roles
- Demonstrate clear task decomposition and agent coordination
- Include appropriate communication protocols between agents

2. Agent Specialization

→ Literature Search Agent: Finds relevant academic papers and sources

→ Analysis Agent: Extracts key insights and data from sources

→ Synthesis Agent: Combines findings into coherent research summaries

→ (Optional: Add additional agents like Citation Verification or Quality Assessment)

3. Communication and Coordination

- Implement structured information sharing between agents
- Include context preservation across agent handoffs
- Demonstrate how agents build on each other's work

4. Governance and Safety

- Include basic safety constraints (e.g., source credibility checks)
- Implement simple conflict resolution mechanisms
- Add monitoring/logging for key agent interactions

5. Working Demonstration

- Process a sample research query end-to-end
- Show how the system decomposes the task across agents
- Produce a coherent output that demonstrates agent collaboration

Submission Output

Submit your prototype including

- Source code with clear documentation
- Architecture diagram showing agent roles and interactions
- Sample output demonstrating the system working on a research query
- Brief reflection on design choices and potential improvements

Technical Notes

- You can use mock APIs or simplified implementations for external services
- Focus on demonstrating multi-agent principles rather than perfect functionality
- Include comments explaining your design decisions
- Consider using frameworks' built-in examples as starting points

Conclusion

After submitting your prototype, you'll engage in a technical review conversation where you'll discuss your implementation choices, identify potential improvements, and explore how your system might scale to handle more complex scenarios."